Replication and reproduction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask whether the artifact, procedure or case population actually changed.

Teach reproduction and replication with honest supplied-fixture/actual-work status.

Keep one habit: Model repeated IDs → compute original/new scores → inspect mismatch candidates → assess new-case accounting → preserve protocol and independence assumptions.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **replication** and reproduction

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of methods results and limitations

Reference example, not the only acceptable wording: Teacher explanation: Methods state what was done; results state observed outcomes; limitations state restrictions on interpretation. Mixing them hides missing **evidence**.

Classify each statement by role and link results to the procedure and denominator. A proposed **mechanism** is an explanation to test, not an extra **observation**.

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of reading a research claim

Reference example, not the only acceptable wording: Teacher explanation: Reading a claim requires its method, data, comparison and uncertainty. Titles and abstracts may omit conditions crucial to interpretation.

Separate the claim from supplied **evidence**. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those **observation**s.

Pictured end goal

Explain **replication** and reproduction using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Methods results and limitations

Reading a research claim

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Reproducing a computation uses recorded artifacts; **replication** collects new **evidence** under a comparable question. Naming conventions vary, so define the terms in the project.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **replication** and reproduction

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach replay value without new-case or independence inflation

> Define terminology locally and inspect the unit, artifact and collection protocol.

Definitions and complete fictional accounting contract

Reproducing a computation uses recorded artifacts; **replication** collects new **evidence** under a comparable question. Naming conventions vary, so define the terms in the project.

Here **replay** means same artifacts/procedure; new-data **replication** means newly collected eligible cases under the same plan. Preserve environment, data versions and raw results.

Term or rule	Complete classroom convention
Replay/reproduce computation	Same saved records and stated analysis procedure/environment
New-data replication	Newly collected eligible cases under a comparable question/protocol; here only supplied fictional new-data fixtures
Main accounting	Original eight IDs Q1–Q8 plus their replay =16 evaluation events, 8 distinct cases, 0 newly added cases
Record identity	Use case IDs, not whether two correctness values happen to be equal
Picture outcome meaning	Binary1=correct, 0=incorrect; correctness flags, not class labels
Picture original/replay	OLD1–OLD4 flags [1, 0, 1, 1]; same artifact/analysis gives 3/4=75%
Picture new-data fixture	NEW1–NEW4 flags [1, 1, 0, 0]; same scoring gives 2/4=50%
Independence	A collection-design assumption requiring scrutiny; distinct IDs alone do not establish it
Evidence status	No actual partner rerun, new collection or model study performed; all records are supplied practice fixtures

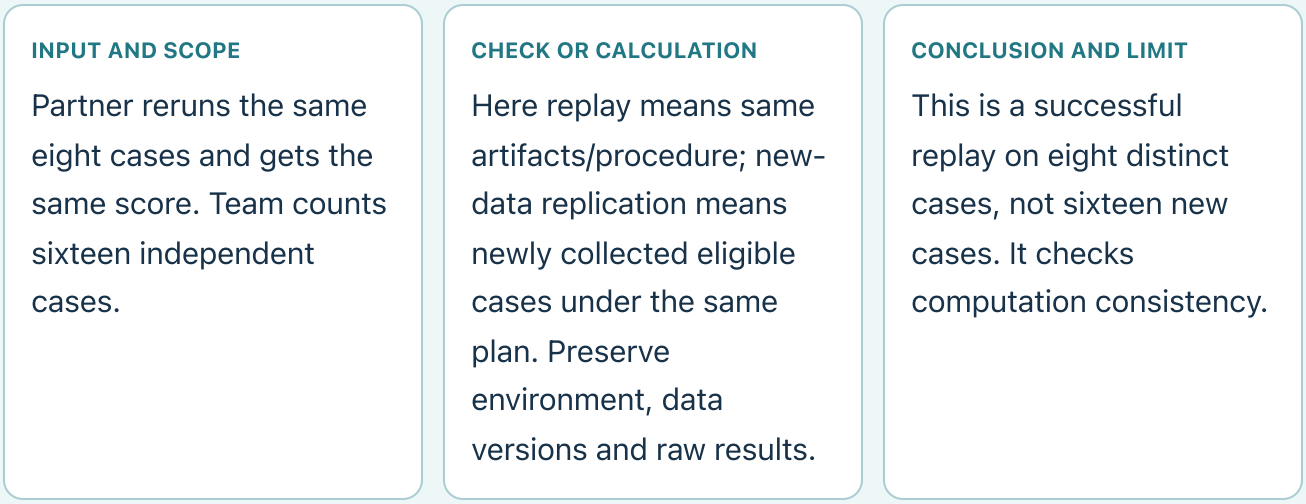
Main event/case ledger and checked reasoning

Main activity	Evaluation events added	Distinct case IDs after activity	What is learned
Original Q1–Q8 evaluated	8	8	Recorded outcome on those cases
Partner scenario reruns Q1–Q8	8	8	Same-score replay checks computational consistency
Main total	16	8	No new-case coverage added by replay
Claim of16 independent cases	Counts repeated events as new units	Unsupported	Neither distinctness nor independence doubled
No actual partner in local code	Only supplied IDs/replay calculation	No human outcome asserted	Record actual collaborator replay only if performed

Main original/replay eight IDs means 16 events, 8 distinct cases and 0 new cases, not 16 independent cases. Distinct IDs do not establish statistical independence either. Picture original flags 3/4=75% and same deterministic replay 75%; new supplied flags 2/4=50%. Across three runs 12 events, 8 distinct IDs, 4 new IDs.

Mismatched original rerun 2/4 needs artifact/analysis/environment investigation; changed-analysis bug is one illustration, not proven cause. A different new-sample result does not automatically imply code error. No actual partner or collection performed in the local fixture. Selected six old plus four genuinely new queries gives 10 distinct total, 4 new; replaying old six does not add coverage.

Replication and reproduction — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: This is a successful **replay** on eight distinct cases, not sixteen new cases. It checks computation consistency.

Qualification: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases.

This is a successful **replay** on eight distinct cases, not sixteen new cases. It checks computation consistency.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Identity, mismatch and evidence-status probes

Replay/replication probe	Expected interpretation
Same old cases rerun	Additional evaluation events, not new distinct cases
Different case IDs	Distinct identifiers, not automatic statistical independence
Different cases share same binary outcome	Still different cases; do not deduplicate by outcome value
Original deterministic records produce different score	Inspect artifact/code/environment/procedure mismatch
New eligible cases produce different score	May reflect sample variation or changed conditions; no automatic code-error conclusion
Same final aggregate score	Useful agreement but not proof every intermediate record/procedure matches
One known replay passes	Does not establish broader reliability or new-data replication
No collaborator actually reruns	Do not invent partner evidence
New fixture supplied rather than collected	Label constructed new-data example, not completed collection study

Separate correctness fixtures and full run accounting

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

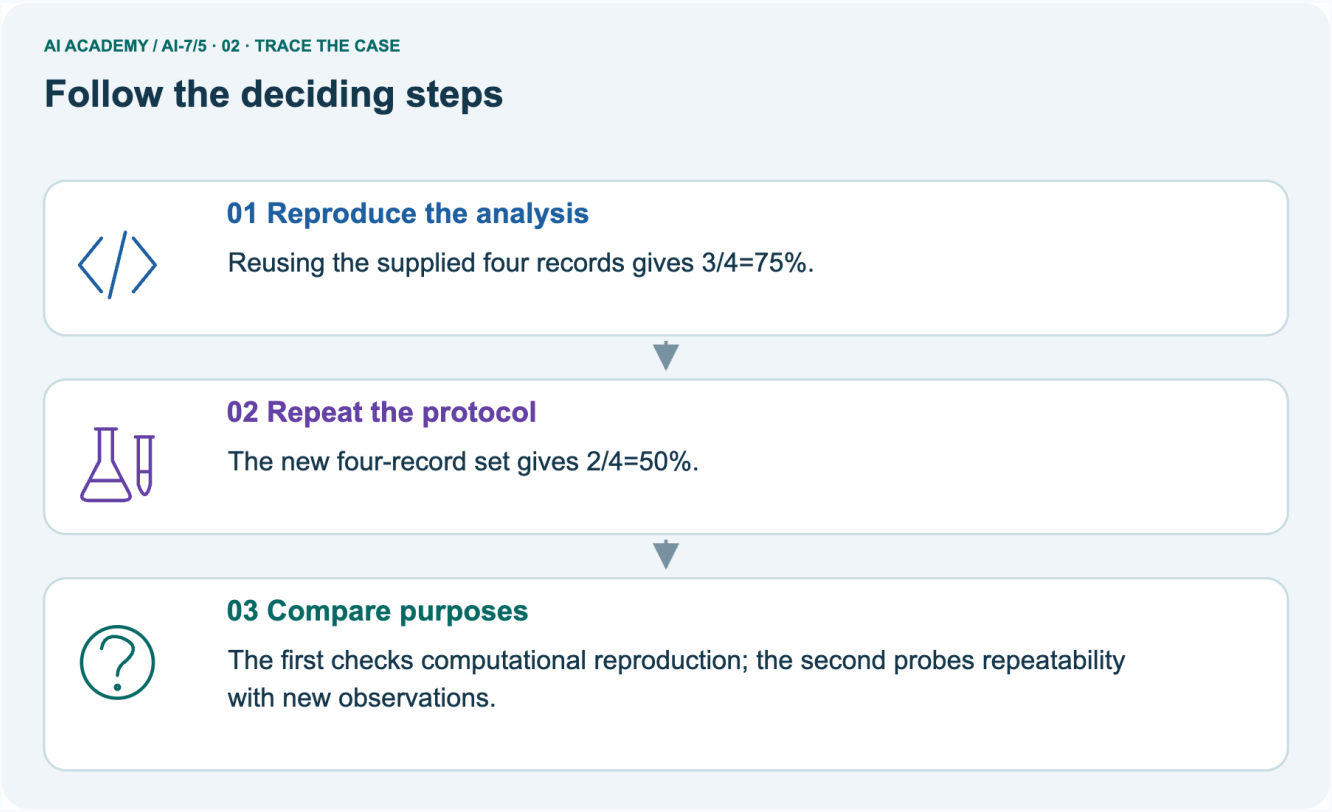
Original recorded correctness flags: [1,0,1,1]. Analysis counts correct cases. New independent set: [1,1,0,0]. Binary entries mean 1=correct and 0=incorrect.

The picture is a separate four-record example. Original correctness flags [1, 0, 1, 1] contain three correct outcomes, so the stated count/total analysis gives 75%. Replaying those same four records with the same deterministic analysis also gives 75%. New supplied records [1, 1, 0, 0] give 50% under that rule. A different new-sample score can arise without a code defect.

But a reported rerun of the original artifact giving 2/4 conflicts with the stated computation: inspect data/version changes, analysis, settings and transcription. The added code demonstrates one deliberately changed analysis as a possible mechanism, not the proven cause of the picture's hypothetical mismatch. The picture assumes newly independent observations; distinct IDs in a fixture do not verify that assumption.

Separate picture activity	Correctness flags	Score	Case accounting
Original saved analysis	1, 0, 1, 1	3/4=75%	4 original IDs
Replay same artifact/procedure	1, 0, 1, 1	3/4=75%	4 more events, 0 new IDs
New-data supplied fixture	1, 1, 0, 0	2/4=50%	4 new IDs
Across all three runs	12 evaluation events	Keep scores separate	8 distinct IDs, 4 newly added
Changed original rerun claims 2/4	Does not match supplied count rule	Investigate mismatch	Do not relabel as new-sample variability

Picture trace and changed-original-rerun key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Reproduce the analysis	Reusing the supplied four records gives 3/4=75%.
2. Repeat the protocol	The new four-record set gives 2/4=50%.
3. Compare purposes	The first checks computational reproduction; the second probes repeatability with new observations.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

Four-case samples are illustrative; neither result alone establishes broad reliability.

Change: The rerun of the original records **output**s 2/4.

Prediction: Investigate code, version, **input** or scoring differences.

Reason: The same fixed records should give the same deterministic count.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Replay	Repeating a computation with the recorded artifacts
Replication	A new evidence collection under a comparable declared question
Distinct unit	A separately eligible observation rather than a repeated copy

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

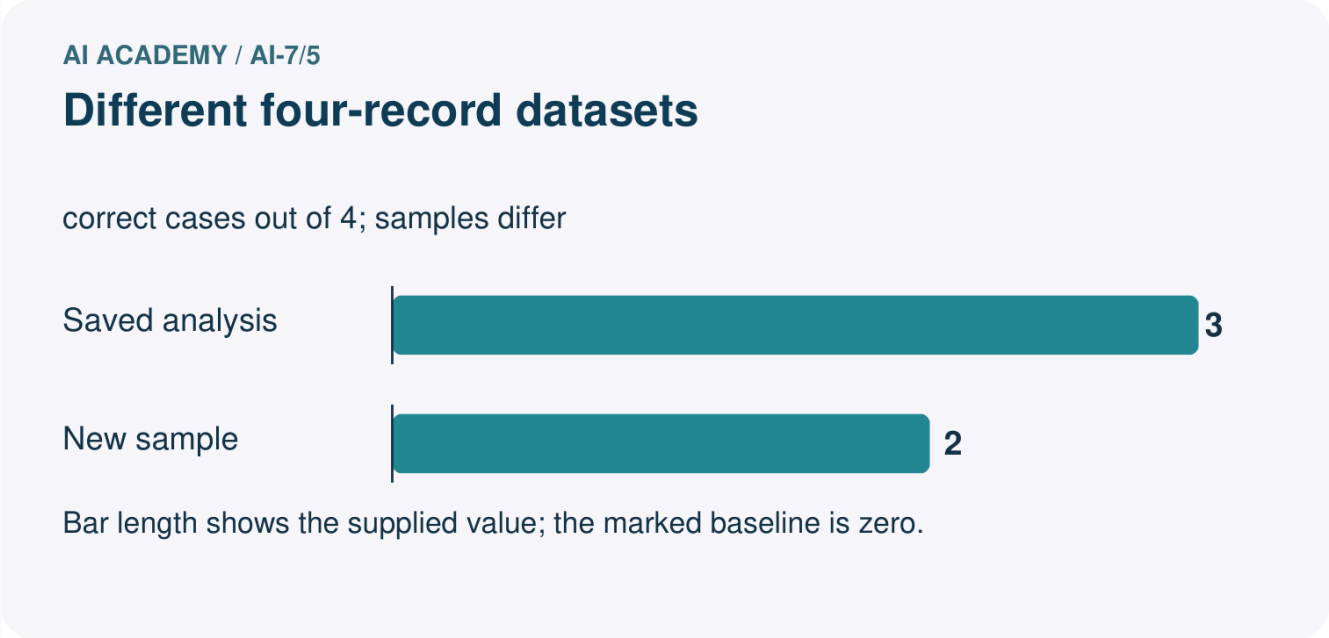
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Replication and reproduction

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/5

Replication and reproduction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Reproduce the analysis

Reusing the supplied four records gives 3/4=75%.



2. Repeat the protocol

The new four-record set gives 2/4=50%.



3. Compare purposes

The first checks computational reproduction; the second probes repeatability with new observations.

CHANGE ONE THING

The rerun of the original records outputs 2/4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: This is a successful **replay** on eight distinct cases, not sixteen new cases. It checks computation consistency.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Replication** and reproduction. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

Reusing the supplied four records gives 3/4=75%. The new four-record set gives 2/4=50%. The first checks computational reproduction; the second probes repeatability with new **observation**s.

Changed case: Investigate code, version, **input** or scoring differences. The same fixed records should give the same deterministic count.

Changed-case reasoning: Investigate code, version, **input** or scoring differences. The same fixed records should give the same deterministic count.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main eight-case replay adds evaluation events but no new distinct cases. Picture original/replay 75% versus supplied new 50% answer different questions; unexpected original 2/4 requires investigation, not automatic new-sample interpretation. IDs alone do not prove independence. Selected old six plus new four gives 10 distinct cases and 4 new; repeated old six are not extra population coverage.

Keep actual collaborator/new-collection work distinct from supplied local fixtures.

Diagnose the current misconceptions

The worked result establishes every future case: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A **replay** checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new **evidence**?

Teacher explanation: Ten distinct total; four are new. **Replay**s of six old cases are not additional **observation**s for population coverage.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Reproducing a computation uses recorded artifacts; **replication** collects new **evidence** under a comparable question. Naming conventions vary, so define the terms in the project.

Here **replay** means same artifacts/procedure; new-data **replication** means newly collected eligible cases under the same plan. Preserve environment, data versions and raw results.

W2 Trace the worked case

Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: This is a successful **replay** on eight distinct cases, not sixteen new cases. It checks computation consistency.

W3 Audit the limit

A report omits this qualification: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: This is a successful **replay** on eight distinct cases, not sixteen new cases. It checks computation consistency.

Qualification: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new **evidence**?

Teacher explanation: Ten distinct total; four are new. **Replay**s of six old cases are not additional **observation**s for population coverage.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Ten distinct total; four are new. **Replay**s of six old cases are not additional **observation**s for population coverage. Require intermediate reasoning and the stated scope.

Fresh retry: Old dataset ten rows rerun three times, new set five rows. Expected reasoning: Fifteen distinct rows; repeated old runs may study run variability but do not add new cases.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **replication** and reproduction **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **replication** and reproduction.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: A literature comparison matrix. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/6, ai-7/8, ai-7/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **replication** and reproduction

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: This is a successful **replay** on eight distinct cases, not sixteen new cases. It checks computation consistency.

Check every learner before discussion: Use the worked case for **Replication** and reproduction. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The rerun of the original records **output**s 2/4. Require a **prediction** and a reason before comparing. Expected result: Investigate code, version, **input** or scoring differences.

Reason: The same fixed records should give the same deterministic count.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

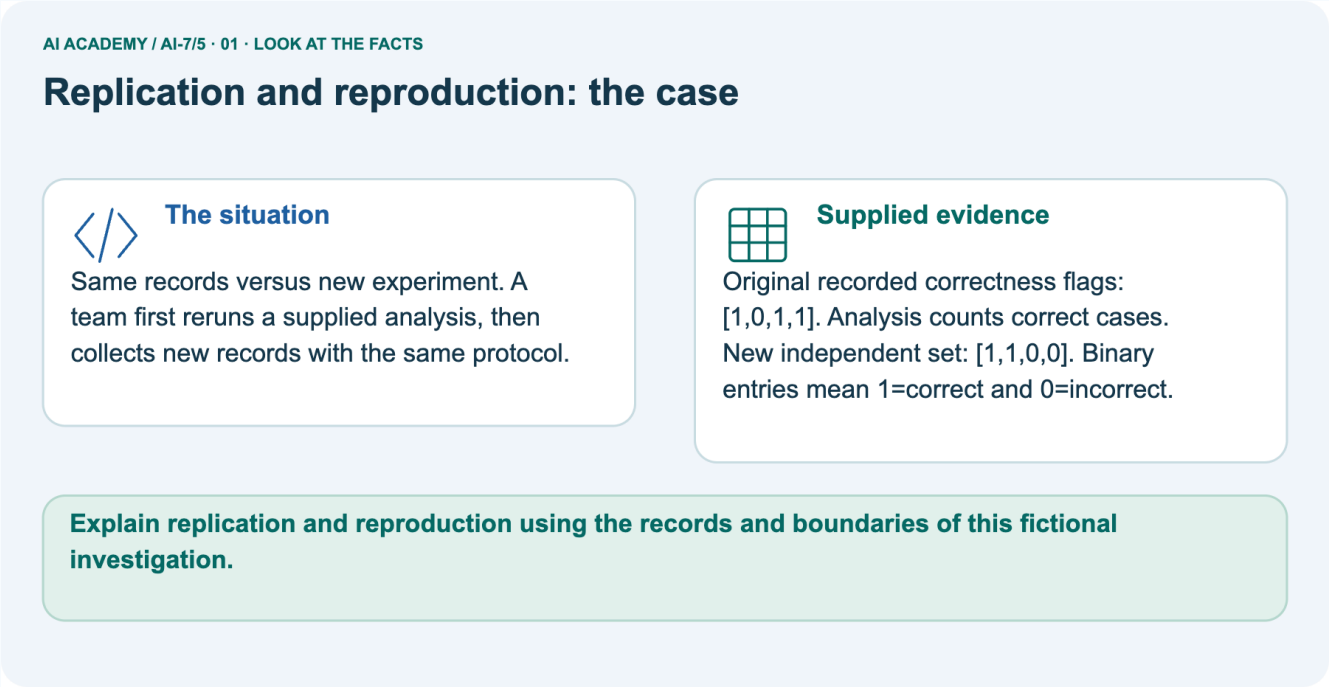
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Replication and reproduction: the case



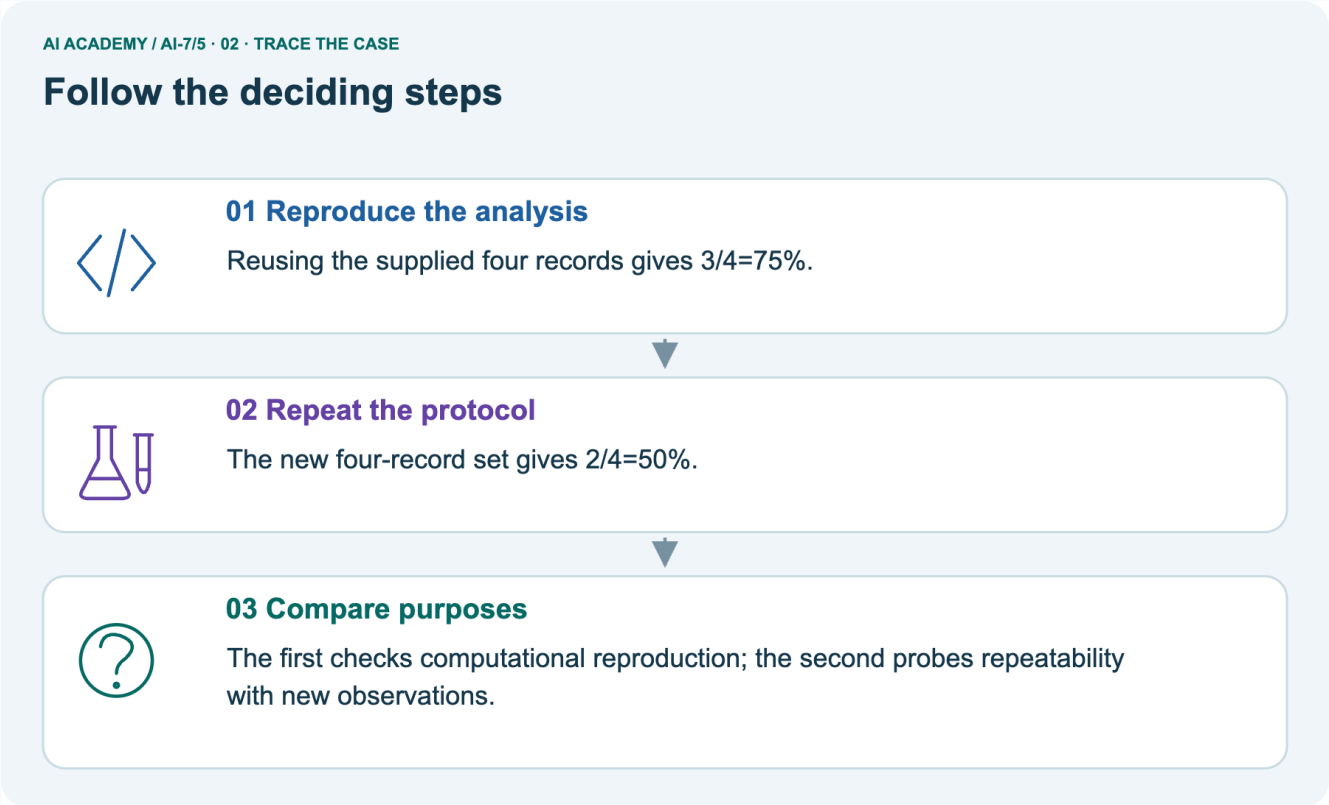
Follow the labelled picture with the matching explanation. Open the vector diagram

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol. Supplied facts: Original recorded correctness flags: [1,0,1,1]. Analysis counts correct cases. New independent set: [1,1,0,0]. Binary entries mean 1=correct and 0=incorrect.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



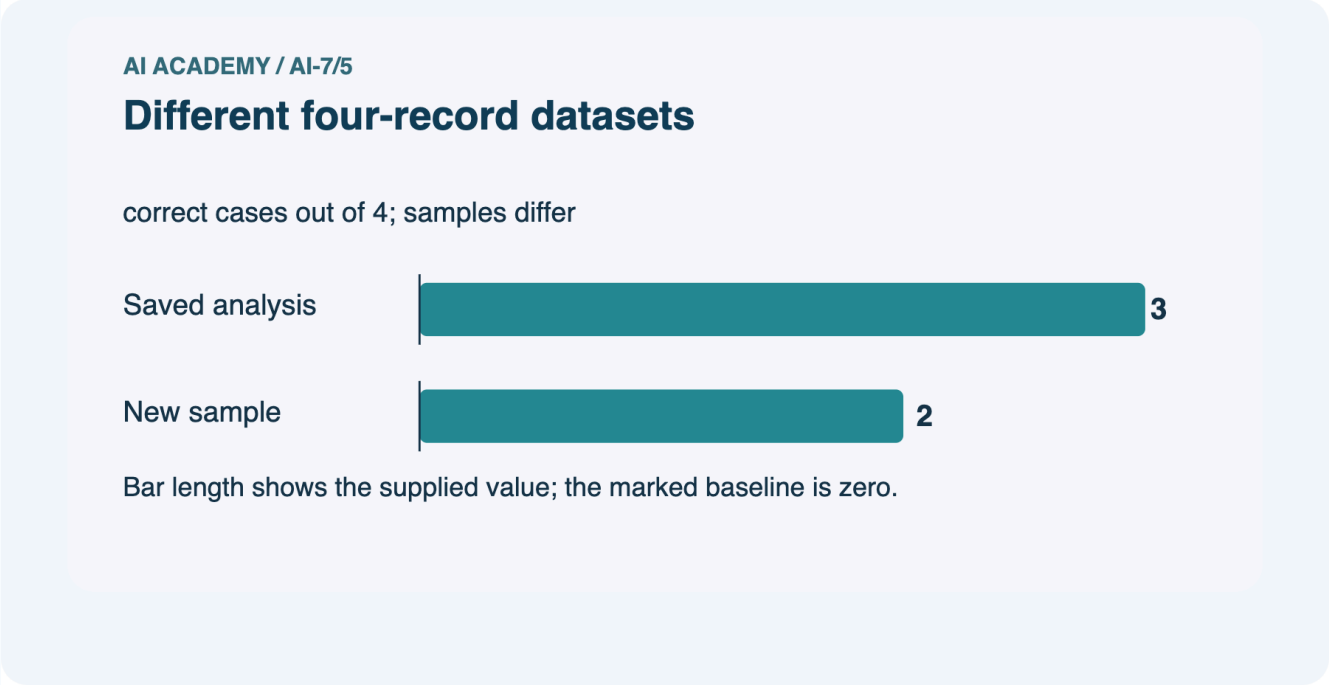
Follow the labelled picture with the matching explanation. Open the vector diagram

Reproduce the analysis: Reusing the supplied four records gives 3/4=75%. → Repeat the protocol: The new four-record set gives 2/4=50%. → Compare purposes: The first checks computational reproduction; the second probes repeatability with new **observation**s.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Different four-record datasets



Follow the labelled picture with the matching explanation. Open the vector diagram

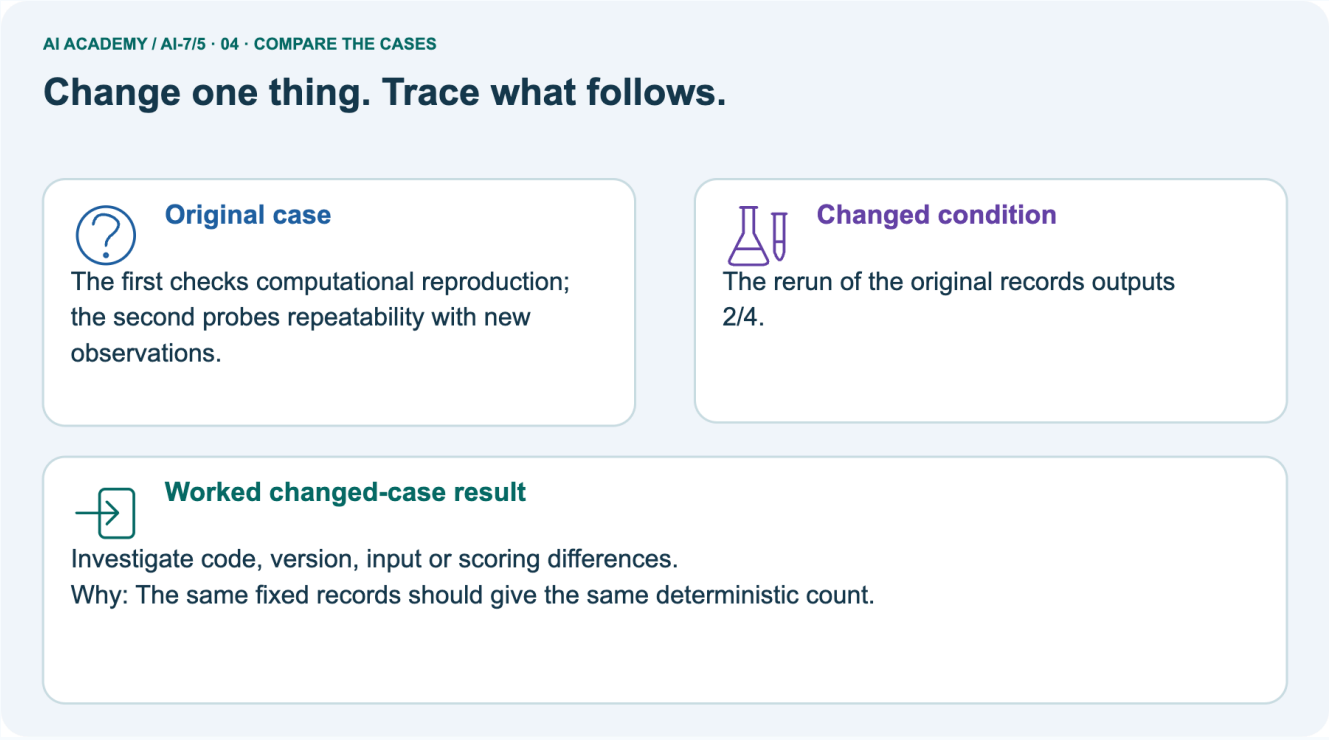
Different four-record datasets. Units: correct cases out of 4; samples differ. Saved analysis: 3. New sample: 2.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The first checks computational reproduction; the second probes repeatability with new **observation**s. Changed condition: The rerun of the original records **output**s 2/4.

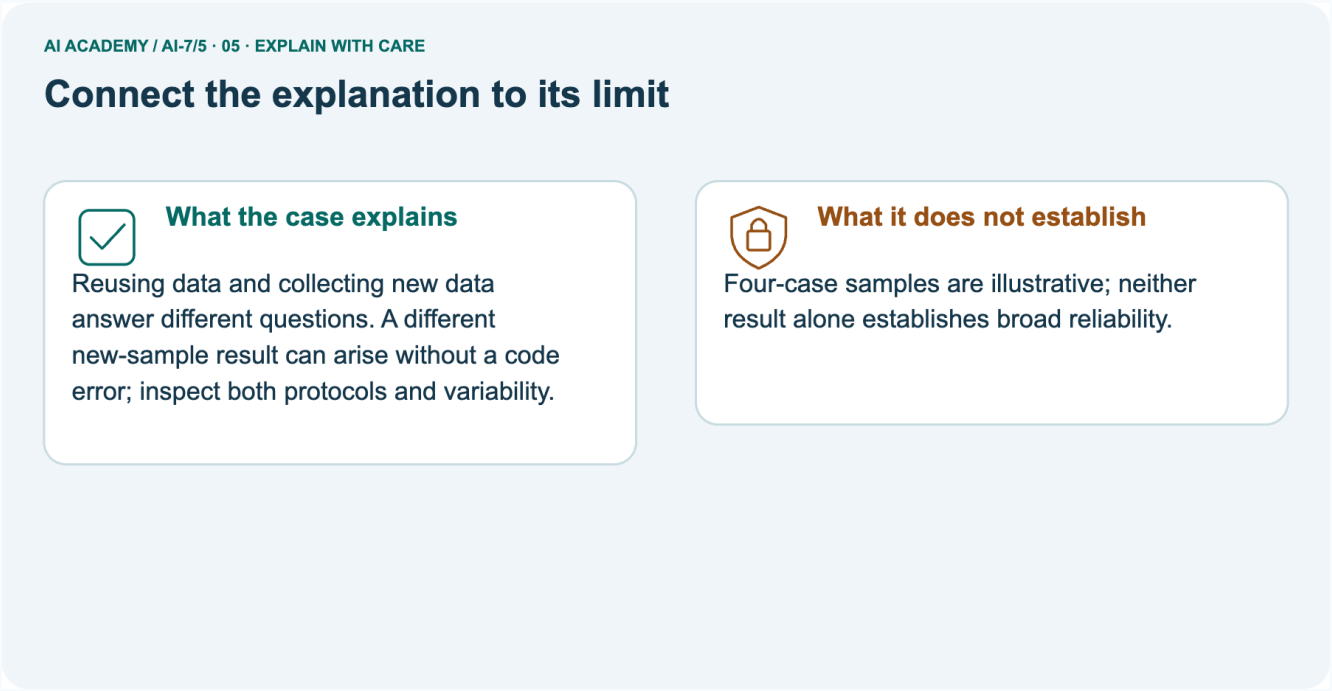
Expected result: Investigate code, version, **input** or scoring differences. Reason: The same fixed records should give the same deterministic count.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The same fixed records should give the same deterministic count.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

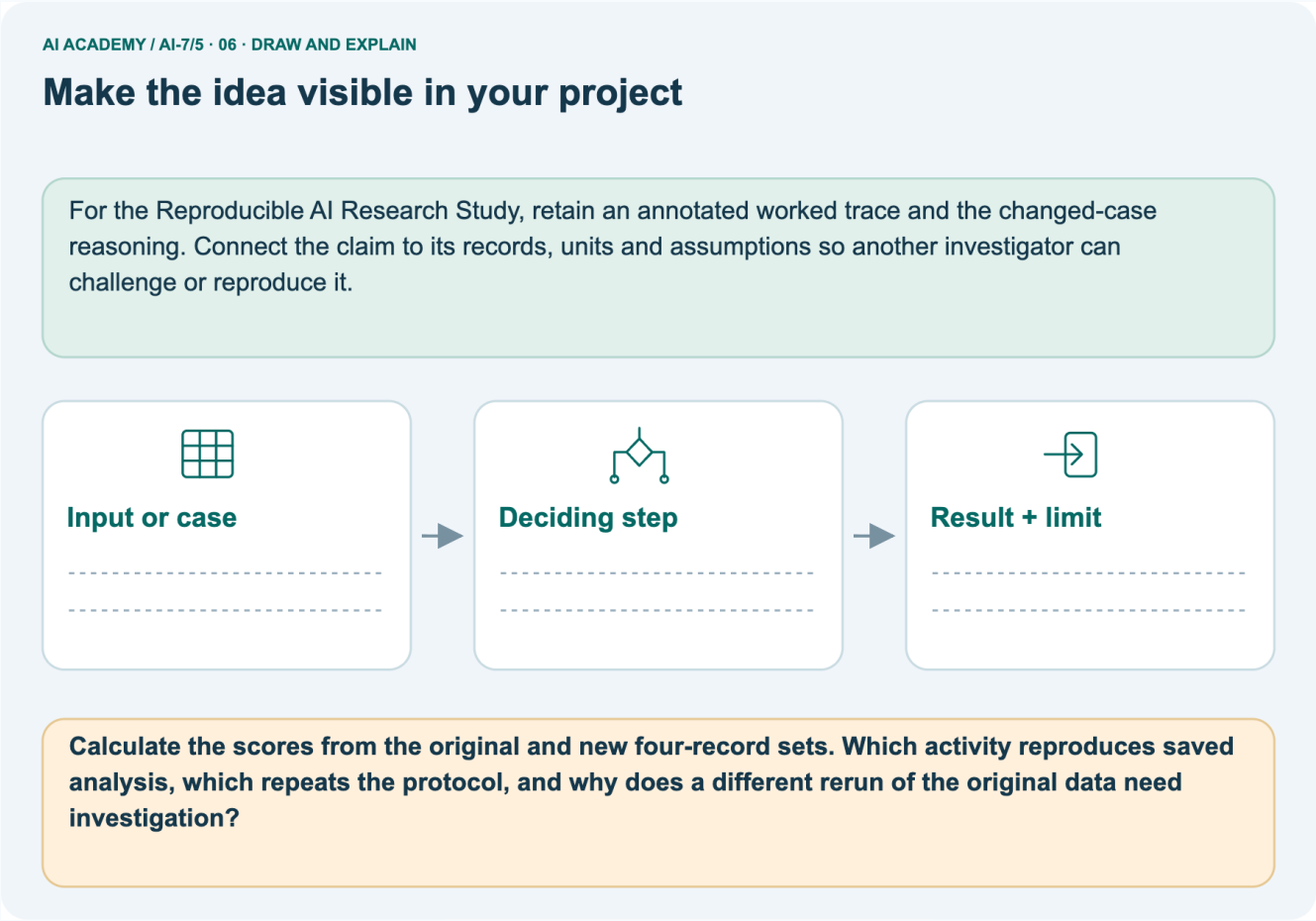
Limit: Four-case samples are illustrative; neither result alone establishes broad reliability.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Four-case samples are illustrative; neither result alone establishes broad reliability.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Reusing the supplied four records gives 3/4=75%. The new four-record set gives 2/4=50%. The first checks computational reproduction; the second probes repeatability with new **observation**s.

Changed case: Investigate code, version, **input** or scoring differences. The same fixed records should give the same deterministic count.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Replication and reproduction: the case

02. Follow the deciding steps

03. Different four-record datasets

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.